FIXED POLYOMINOES WITH n CELLS AND WIDTH $n-2$: A FORMAL PROOF OF $T(n, n-2) = 8n^2 - 51n + 86$ (OEIS A308359)

MATH0

Abstract. Let $T(n, w)$ be the number of fixed polyominoes with n cells whose bounding box has width w (OEIS A308359). We prove that $T(n, n-2) = 8n^2 - 51n + 86$ for every $n \geq 5$, which R. J. Mathar recorded as a conjecture in the entry in 2019. The formula can also be derived from two theorems of Goupil, Cloutier and Nouboud (2010) on polyominoes inscribed in a rectangle, so we classify our result as the first formal proof of a known result. Our proof is self-contained: it shows that such a polyomino lies in a strip of height 3 and has interval columns, and counts these shapes with a 36-state transfer matrix, whose counting vector has quadratic polynomials in the width as entries. It is verified in Lean 4 against a statement that math0 formalized from the OEIS entry, reviewed for fidelity, and frozen before the proof attempt.

1. Introduction

A *fixed polyomino* with n cells is a set of n unit squares of the square grid, connected through shared edges, taken up to translation only. Their number is OEIS A001168 [6]. The triangle A308359 [8], contributed by R. J. Mathar on May 22, 2019, refines this count: "Triangle T(n,w) read by rows: the number of fixed polyominoes with n cells and width w of the convex hull." Its rows for $n \leq 7$ are $1;\ 1,1;\ 1,4,1;\ 1,9,8,1;\ 1,18,31,12,1;\ 1,35,95,68,16,1;\ 1,66,269,282,121,20,1$. The entry gives $T(n,1) = T(n,n) = 1$ and $T(n,n-1) = 4n-8$ for $n \geq 3$, and its FORMULA section, part of Mathar's entry, states

"`Conjecture: T(n,n-2) = 8*n^2 - 51*n + 86 for n >= 5.`"

The column $w = 3$ is A335606 [9], and A292357 [7] counts fixed polyominoes by width and height. In the current revision of A308359 (#23, October 10, 2022) the formula is still marked as a conjecture, and no proof is recorded. We prove it (Theorem 2).

Prior work. The math0 literature referee observed that the formula follows from results of Goupil, Cloutier and Nouboud [2], published nine years before the conjecture. They count the polyominoes *inscribed* in a $b \times k$ rectangle (that is, with this bounding box) that have the minimum number $b + k - 1$ of cells, $p_{\min}(b,k)$, and those with $b + k$ cells, $p_{\min+1}(b,k)$. A polyomino of width w with $w+2$ cells has height 3 (minimum area) or 2 (minimum area plus one), so $T(w+2, w) = p_{\min}(w,3) + p_{\min+1}(w,2)$, and their Theorems 1 and 3 evaluate this to $8w^2 - 19w + 16$ (Remark 12). This consequence is not stated in [2], which predates the triangle, and the referees found no place where it is drawn: the OEIS entry records no proof, and they found none on arXiv, MathOverflow, Mathematics Stack Exchange or the SeqFan list. Marin [3] constructs automata for polyominoes inscribed in a rectangle of fixed width and growing height; this does not address the diagonal $w = n - 2$. Since the statement follows from published theorems, we classify our result as the *first formal proof of a known result*. Our proof (Section 3) does not port the case analysis of [2]: it reduces the count to a transfer matrix on 36 states and certifies quadratic closed forms by finite computations that the Lean kernel checks.

Date: October 19, 2026.

2. Statement

We write cells as pairs $(x, y) \in \mathbb{N}^2$, with column x and row y. Two cells are *adjacent* if they agree in one coordinate and differ by 1 in the other, and a finite set $S \subseteq \mathbb{N}^2$ is *connected* if any two of its cells are joined by a chain of cells of S in which consecutive cells are adjacent. Each translation class of finite nonempty sets of cells of $\mathbb{Z}^2$ contains exactly one set $S \subseteq \mathbb{N}^2$ that meets both the column $x = 0$ and the row $y = 0$. Its bounding box (equivalently, its convex hull) has width $\max_{p \in S} p_x + 1$.

Definition 1. A *normalized polyomino with n cells and width w* is a connected set $S \subseteq \mathbb{N}^2$ with $|S| = n$ that meets the column $x = 0$ and the row $y = 0$ and has $\max_{p \in S} p_x + 1 = w$. Let $T(n, w)$ be the number of these sets.

Theorem 2 (Mathar's conjecture)**.** *For every integer $n \geq 5$, $T(n, n-2) = 8n^2 - 51n + 86$. Equivalently, $T(w+2, w) = 8w^2 - 19w + 16$ for every $w \geq 3$.*

For $n = 5, \ldots, 11$ the formula gives the listed values $31, 68, 121, 190, 275, 376, 493$. For $w = 1, 2$ it fails: $T(3, 1) = 1$ and $T(4, 2) = 9$, while $8w^2 - 19w + 16$ equals 5 and 10.

The Lean statement. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A308359.lean`, reviewed for fidelity against the OEIS entry, and frozen (SHA-256 `84eea4b2...80f7`) before any proof attempt. Its definitions (without their docstrings and without the `DecidableRel` instance for `Adj`) and the statement (with its docstring) read verbatim, in the namespace `Math0.Statements.A308359`:

```
def Adj (p q : ℕ × ℕ) : Prop :=
  (p.1 = q.1 ∧ (p.2 + 1 = q.2 ∨ q.2 + 1 = p.2)) ∨ (p.2 = q.2 ∧ (p.1 + 1 = q.1 ∨ q.1 + 1 = p.1))
def Step (S : Finset (ℕ × ℕ)) (p q : ℕ × ℕ) : Prop := p ∈ S ∧ q ∈ S ∧ Adj p q
def IsConnected (S : Finset (ℕ × ℕ)) : Prop :=
  ∀ p ∈ S, ∀ q ∈ S, Relation.ReflTransGen (Step S) p q
def IsPoly (n w : ℕ) (S : Finset (ℕ × ℕ)) : Prop :=
  S.card = n ∧ (∃ p ∈ S, p.1 = 0) ∧ (∃ p ∈ S, p.2 = 0) ∧ S.sup Prod.fst + 1 = w ∧ IsConnected S
noncomputable def T (n w : ℕ) : ℕ := {S : Finset (ℕ × ℕ) | IsPoly n w S}.ncard

/-- Conjecture (Mathar, A308359): for every `n ≥ 5`, the number of fixed polyominoes with `n`
cells and width `n - 2` is `T(n, n - 2) = 8n^2 - 51n + 86`. -/
def statement : Prop :=
  ∀ n : ℕ, 5 ≤ n → (T n (n - 2) : ℤ) = 8 * (n : ℤ) ^ 2 - 51 * n + 86
```

How the Lean definitions encode the entry. `IsPoly n w S` is Definition 1: `IsConnected` is connectivity through chains of adjacent cells (the reflexive–transitive closure of `Step S`), and `S.sup Prod.fst` is $\max_{p \in S} p_x$. `T` counts with `Set.ncard`, which returns 0 on infinite sets; this junk value does not arise, since the statement file proves (`T_eq_Tbox`) that every such S lies in the box $[0, w) \times [0, n + 1 - w)$ (Lemma 5), so that $T(n, w)$ equals a computable count `Tbox n w` over the n-element subsets of the box. For $n \geq 5$ the truncated subtraction $n - 2$ in $\mathbb{N}$ is exact, and the right-hand side is computed in $\mathbb{Z}$.

Only the width is prescribed. The comment line of the entry says that the height h satisfies $w \leq h \leq n$, but the entry contradicts this: $T(4, 3) = 8$ counts the eight tetrominoes of width 3, all of height 2; the formula $T(n, n-1) = 4n - 8$ is annotated "width n-1 and height 2"; and the row sums are A001168. Under the restriction $h \geq w$ the listed $T(6, 4) = 68$ would be 0, since a polyomino of width 4 with 6 cells has height at most 3 (Lemma 5). The formalization follows the data. To check it against the data, the statement file defines the triangle read by rows, `a`, proves $a(n(n-1)/2 + w) = T(n, w)$ for $1 \leq w \leq n$ (`a_eq_T`), and proves, by four `decide +kernel` computations, that $a(1), \ldots, a(23)$ are the first 23 listed terms (`a_data`: rows 1 to 6, $T(7, 1) = 1$, $T(7, 2) = 66$), including the first two values $T(5, 3) = 31$ and $T(6, 4) = 68$ of

the conjecture. For this evaluation, connectivity is decided by a flood fill on bit masks, proved equivalent to `IsConnected` (`isConnected_iff_bits`).

3. Proof

The *column* x of $S \subseteq \mathbb{N}^2$ is $S_x = \{y : (x, y) \in S\}$, and $[a, b] = \{y \in \mathbb{N} : a \le y \le b\}$. The first three lemmas are proved in the statement file.

Lemma 3 (`frontier`, `exists_coord`)**.** *Let $c_0, \dots, c_m$ be a chain in S. (1) If $c_0 \in A$ and $c_m \notin A$, then some cell of $A \cap S$ is adjacent to some cell of $S \setminus A$. (2) If f is one of the two coordinates, every integer k between $f(c_0)$ and $f(c_m)$ equals $f(c)$ for some $c \in S$.*

Proof. (1) If j is the first index with $c_j \notin A$, then $c_{j-1} \in A$ is adjacent to c_j. (2) Each step changes f by at most 1. □

Lemma 4 (`cols_add_rows_le`)**.** *If S is nonempty, connected, and occupies C columns and R rows, then $C + R \le |S| + 1$.*

Proof. Start from one cell, which occupies $1 + 1$ columns and rows. While some cell of S is not yet taken, Lemma 3(1) (applied to a chain from a taken cell to an untaken one) gives an untaken cell c adjacent to a taken cell b. Take c. As c shares its column or its row with b, the number of occupied columns and rows grows by at most 1 per cell. □

Lemma 5 (`subset_box`)**.** *A normalized polyomino S with n cells and width w lies in $[0, w) \times [0, n + 1 - w)$, and its columns $S_0, \dots, S_{w-1}$ are nonempty.*

Proof. Lemma 3(2), applied to a chain from a cell in column 0 to a cell in column $w - 1$, shows that all w columns $0, \dots, w - 1$ are occupied. For $p \in S$, a chain from a cell in row 0 to p shows that the rows $0, \dots, p_y$ are occupied. By Lemma 4, $w + p_y + 1 \le n + 1$. Finally $p_x \le w - 1$ by the definition of the width. □

From now on $w \ge 1$ and $n = w + 2$. By Lemma 5, a normalized polyomino with $w + 2$ cells and width w lies in $[0, w) \times \{0, 1, 2\}$ and has w nonempty columns, so

$$\sum_{x<w} \big(|S_x| - 1\big) = |S| - w = 2. \tag{1}$$

Definition 6 (`Good`)**.** A finite set $S \subseteq \mathbb{N}^2$ is *good of width w* if (i) all cells of S lie in columns $x < w$; (ii) for every $x < w$, $S_x = [a_x, b_x]$ for some $a_x \le b_x \le 2$; and (iii) for every x with $x + 1 < w$, the columns S_x and S_{x+1} share a row.

Lemma 7 (`good_conn`)**.** *Every good set is connected.*

Proof. Let S be good of width $w \ge 1$ (for $w = 0$, $S = \emptyset$). By induction on $x < w$, every cell of column x is joined to $(0, a_0)$ by a chain in S: for $x = 0$, walk vertically inside the interval S_0; for $x + 1$, take a row y shared by S_x and S_{x+1}, reach (x, y) by induction, step to $(x + 1, y)$, and walk vertically inside S_{x+1}. Since adjacency is symmetric, any two cells are joined through $(0, a_0)$. □

Lemma 8 (`cross`, `no_gap`)**.** *Let S be connected, with $|S| = w + 2$, all cells in columns $< w$, and every column $x < w$ nonempty. Then no column S_x contains the rows 0 and 2 but not the row 1.*

Proof. A chain from a row $\le k$ to a row $> k$ contains a vertical step from some (c, k) to $(c, k + 1)$, namely its first step into a row $> k$ (`cross`). Suppose $(x, 0), (x, 2) \in S$ and $(x, 1) \notin S$. Applying this to a chain from $(x, 0)$ to $(x, 2)$ with $k = 0$ and with $k = 1$ gives columns c_1, c_2 with $\{0, 1\} \subseteq S_{c_1}$ and $\{1, 2\} \subseteq S_{c_2}$, both different from x because $1 \notin S_x$. If $c_1 = c_2$, then $|S_{c_1}| - 1 \ge 2$ and $|S_x| - 1 \ge 1$; otherwise x, c_1, c_2 are three distinct columns with $|S.| - 1 \ge 1$. Either way $\sum_{x'<w}(|S_{x'}| - 1) \ge 3$, whereas the hypotheses give $\sum_{x'<w}(|S_{x'}| - 1) = |S| - w = 2$. □

Proposition 9 (`struct`, `poly_iff`). *Let $w \geq 1$. A set $S \subseteq \mathbb{N}^2$ is a normalized polyomino with $w+2$ cells and width w if and only if S is good of width w, $|S| = w+2$, and S meets the row 0.*

Proof. Let S be such a polyomino. By Lemma 5, (i) holds and each S_x, $x < w$, is a nonempty subset of $\{0,1,2\}$; the only such subset that is not an interval is $\{0,2\}$, which Lemma 8 excludes, so (ii) holds. For (iii), suppose S_x and S_{x+1} share no row, and let A be the set of cells of S in columns $\leq x$. By Lemma 3(1), applied to a chain from column x to column $x+1$, some $b \in A$ is adjacent to some $c \in S \setminus A$. Then $b_x \leq x < c_x$, so $b = (x, y)$ and $c = (x+1, y)$ for a row y shared by S_x and S_{x+1}, a contradiction.

Conversely, a good set of width w with $w+2$ cells that meets row 0 is connected (Lemma 7), meets column 0 as $S_0 \neq \emptyset$, and has largest column $w-1$ as $S_{w-1} \neq \emptyset$. □

The transfer matrix. A *state* is a quadruple $s = (e, a, b, z)$ with $e \in \{0,1,2\}$, $0 \leq a \leq b \leq 2$ and $z \in \{\mathtt{t}, \mathtt{f}\}$ (true, false); there are 36 states (`states`). For $w \geq 1$, a good set S of width w is *in state* s (`InP`) if $|S| = w + e$, $S_{w-1} = [a, b]$, and $z = \mathtt{t}$ exactly when S meets row 0. Let $P(w, s)$ be the (finite) set of good sets of width w in state s. A good set is in at most one state (`Pset_unique`). A state $t = (e', a', b', z')$ *feeds* $s = (e, a, b, z)$ (`Compat`) if

$$e = e' + (b - a), \qquad a \leq b' \text{ and } a' \leq b \ \text{ (i.e. } [a,b] \cap [a',b'] \neq \emptyset), \qquad z = (z' \text{ or } a = 0).$$

Define $D(1, s) = 1$ if $e = b - a$ and $z = [a = 0]$, $D(1, s) = 0$ otherwise, and $D(w+1, s) = \sum_{t \text{ feeds } s} D(w, t)$ for $w \geq 1$.

Lemma 10 (`Pset_one`, `Pset_succ`, `card_Pset`). *For every $w \geq 1$ and every state s, $|P(w, s)| = D(w, s)$.*

Proof. A good set of width 1 with last column $[a, b]$ is $\{0\} \times [a, b]$; it has $b - a + 1$ cells and meets row 0 iff $a = 0$. So $|P(1, s)| = D(1, s)$.

Let $w \geq 1$ and $s = (e, a, b, z)$. We show that $S \mapsto S' = \{p \in S : p_x < w\}$ is a bijection from $P(w+1, s)$ onto the union of the sets $P(w, t)$ over the states t feeding s, which is disjoint by uniqueness of states, with inverse $S' \mapsto S' \cup (\{w\} \times [a, b])$. Let $S \in P(w+1, s)$. Then S' is good of width w, so $|S'| \geq w$; write $|S'| = w + e'$, $S'_{w-1} = [a', b']$, and let z' record whether S' meets row 0. From $|S| = |S'| + (b - a + 1)$ we get $e = e' + (b - a)$, so $e' \leq 2$ and $t = (e', a', b', z')$ is a state. Columns $w-1$ and w of S share a row, so $[a, b] \cap [a', b'] \neq \emptyset$; and S meets row 0 iff S' does or $a = 0$. So t feeds s and $S' \in P(w, t)$. Conversely, let $S' \in P(w, t)$ with t feeding s, and $S = S' \cup (\{w\} \times [a, b])$. Then (i) and (ii) hold for width $w+1$; for (iii), the row $\max(a, a')$ lies in $[a', b'] \cap [a, b]$. Moreover $|S| = w + e' + (b - a + 1) = (w+1) + e$, $S_w = [a, b]$, and S meets row 0 iff z' or $a = 0$, that is, iff z. So $S \in P(w+1, s)$. The two maps are mutually inverse. Hence $|P(w+1, s)| = \sum_{t \text{ feeds } s} |P(w, t)|$, and the lemma follows by induction on w. □

Lemma 11 (`D_three`, `coefA`, `coefB`, `coefC`, `D_poly`). *With $(A_s, B_s, C_s) \in \mathbb{Z}^3$ as in Table 1, $D(w, s) = A_s w^2 + B_s w + C_s$ for every $w \geq 3$ and every state s.*

Proof. Induction on $w \geq 3$. For $w = 3$ the identity is checked for all 36 states by evaluation. If it holds for w, then $D(w+1, s) = \sum_{t \text{ feeds } s} (A_t w^2 + B_t w + C_t)$, while $A_s(w+1)^2 + B_s(w+1) + C_s = A_s w^2 + (2A_s + B_s)w + (A_s + B_s + C_s)$. So it suffices that for each of the 36 states s

$$\sum_{t \text{ feeds } s} A_t = A_s, \qquad \sum_{t \text{ feeds } s} B_t = 2A_s + B_s, \qquad \sum_{t \text{ feeds } s} C_t = A_s + B_s + C_s. \tag{2}$$

These 108 integer identities are checked by evaluation. For example, the states feeding $s = (2, 2, 2, \mathtt{t})$ have $e' = 2$, $b' = 2$, $z' = \mathtt{t}$; they are $(2, a', 2, \mathtt{t})$ for $a' = 0, 1, 2$, and the sums are $(0 + 0 + 1,\ 0 + 2 - 1,\ 3 - 3 - 1) = (1, 1, -1) = (A_s,\ 2A_s + B_s,\ A_s + B_s + C_s)$. □

s	$D(w,s)$	s	$D(w,s)$	s	$D(w,s)$
$(0,0,0,\mathtt{t})$	1	$(1,1,1,\mathtt{t})$	$2w-3$	$(2,1,1,\mathtt{f})$	$2w^2-10w+13$
$(0,1,1,\mathtt{f})$	1	$(1,1,2,\mathtt{f})$	2	$(2,1,1,\mathtt{t})$	$4w^2-15w+16$
$(0,2,2,\mathtt{f})$	1	$(1,2,2,\mathtt{f})$	$2w-3$	$(2,1,2,\mathtt{f})$	$4w-8$
$(1,0,0,\mathtt{t})$	$2w-3$	$(2,0,0,\mathtt{t})$	$3w^2-11w+12$	$(2,1,2,\mathtt{t})$	$2w-3$
$(1,0,1,\mathtt{t})$	2	$(2,0,1,\mathtt{t})$	$6w-11$	$(2,2,2,\mathtt{f})$	$2w^2-10w+13$
$(1,1,1,\mathtt{f})$	$2w-3$	$(2,0,2,\mathtt{t})$	3	$(2,2,2,\mathtt{t})$	w^2-w-1

TABLE 1. The closed forms $D(w,s) = A_s w^2 + B_s w + C_s$, $w \geq 3$ (the Lean function `coef`). For the other 18 states, $(A_s, B_s, C_s) = (0,0,0)$; no good set is in these states (e.g. $(0,0,1,z)$ has a 2-cell column but excess 0, and $(e,0,b,\mathtt{f})$ uses row 0). The six final states are those with $e=2$ and $z=\mathtt{t}$.

Proof of Theorem 2. Let $n \geq 5$, $w = n-2 \geq 3$, and let F be the set of the six final states $(2,a,b,\mathtt{t})$. By Proposition 9, the sets counted by $T(w+2,w)$ are the good sets of width w with $w+2$ cells meeting row 0, that is, the members of the pairwise disjoint sets $P(w,s)$, $s \in F$ (`T_eq`). By Lemmas 10 and 11 and Table 1 (`final_sum`),

$$T(n,n-2) = \sum_{s\in F}\big(A_s w^2 + B_s w + C_s\big) = (3+4+1)w^2 + (-11+6-15+2-1)w + (12-11+3+16-3-1),$$

which is $8w^2 - 19w + 16 = 8n^2 - 51n + 86$. □

Remark 12 (derivation from [2]; not formalized)**.** By Lemma 4, a polyomino of width w and height h has at least $w+h-1$ cells, so with $w+2$ cells $h \leq 3$; and $h=1$ would mean w cells. Those of height 3 are inscribed in a $w \times 3$ rectangle with minimum area $w+2$, and those of height 2 are inscribed in a $w \times 2$ rectangle with area $w+2$, the minimum plus one. A quarter-turn shows that the orientation of the rectangle does not matter. Theorem 1 of [2], $p_{\min}(b,k) = 2k+2b-3bk-8+8\binom{b+k-2}{b-1}$ for $b,k \geq 2$, gives $p_{\min}(w,3) = 4w^2-3w-2$, and Theorem 3 of [2] gives $p_{\min+1}(b,2) = 4b^2-16b+18$ for $b > 2$; the sum is $8w^2-19w+16$ (for $w = 3,4$: $25+6$ and $50+18$). Theorem 1 with $k=2$ also gives the entry's $T(n,n-1) = p_{\min}(n-1,2) = 4n-8$.

4. Machine verification

Files, declarations and kernel computations. The proof, about 750 lines, is the file `lean/Math0/Frontier/OEIS/A308359.lean` of the math0 repository `https://github.com/Foundation-0/math0`. It imports the frozen statement file without modifying it, proves `theorem result : statement` in the namespace `Math0.Frontier.A308359`, and ends with

```
#math0_settles Math0.Frontier.A308359.result Math0.Statements.A308359.statement
```

Section 3 names the declaration behind each step; Lemmas 3–5 are in the statement file, the rest are private declarations of the proof file, where $P(w,s)$ is `Pset w s`, a filter of the powerset of $[0,w) \times [0,3)$. Five finite facts are proved by `decide +kernel`: `D_three` (the base case of Lemma 11), `coefA`, `coefB`, `coefC` (the identities (2) for all 36 states), and `final_sum`. With `decide +kernel` the Lean kernel itself reduces the `Decidable` instance of the proposition to `isTrue`, so these steps add no axioms and trust no compiled code; the first four carry `set_option maxHeartbeats 4000000`, a resource limit. Neither file uses `native_decide`, `sorry`, or a declared axiom.

Audit. `#math0_settles` (in `lean/Math0/Audit.lean`) fails unless (1) `statement` is a `Prop`-valued definition without universe parameters; (2) `statement` and `result` use no axioms besides `propext`, `Classical.choice` and `Quot.sound` (excluding `sorryAx`, the axioms behind `native_decide`, and custom axioms); and (3) the type of `result` is syntactically `statement` or its negation, up to metadata, with no unfolding. It reports

```
math0 audit: 'Math0.Statements.A308359.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A308359.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A308359.result' proves 'Math0.Statements.A308359.statement'
```

Toolchain and reproduction. We use Lean 4 [5] `v4.33.1` and Mathlib [4] `v4.33.1`, pinned in `lean/`. The repository also pins formal-conjectures [1] at commit `c77fdd4`; this conjecture is not in that collection, so math0 formalized it. To reproduce, run `scripts/verify.sh` from the repository root: it builds all math0 results with `lake build`, re-running every audit command, and checks that the pinned dependencies are unmodified. The single file is checked by `lake env lean Math0/Frontier/OEIS/A308359.lean` in `lean/`. Independently of the formal proof, the referees checked the formula numerically: an enumeration of all fixed polyominoes with up to 13 cells agreed with the listed terms, and a column transfer matrix with full connectivity tracking, not using Proposition 9, confirmed it for $5 \le n \le 32$.

References

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[6] The OEIS Foundation Inc., *Number of fixed polyominoes with n cells*, `https://oeis.org/A001168`.

[7] The OEIS Foundation Inc., `https://oeis.org/A292357`, entry by A. Howroyd (Oct. 2, 2017).

[8] The OEIS Foundation Inc., `https://oeis.org/A308359`, entry and conjecture by R. J. Mathar (May 22, 2019); revision #23, Oct. 10, 2022.

[9] The OEIS Foundation Inc., `https://oeis.org/A335606`, entry by R. J. Mathar (Jun. 15, 2020).

Produced by Claude (an AI model developed by Anthropic) in the math0 project